

Foundation (Jalapeno)

Q1. Solve the inequality $2x + 5 > 11$.

- (A) $x > 3$
- (B) $x < 3$
- (C) $x > -3$
- (D) $x < -3$
- (E) $x = 3$

Q2. A basketball player scores x points per game. If she scores more than 15 points per game, which inequality represents this situation?

- (A) $x < 15$
- (B) $x > 15$
- (C) $x = 15$
- (D) $x \leq 15$
- (E) $x \geq 15$

Q3. Graph the solution to $x - 2 < 7$ on a number line.

- (A) $(-\infty, 9)$
- (B) $(-\infty, 5)$
- (C) $(-\infty, 10)$
- (D) $[9, \infty)$
- (E) $[5, \infty)$

Q4. A cyclist rides x miles in y hours. If she rides more than 20 miles in 2 hours, which inequality represents this situation?

- (A) $x < 20$
- (B) $x > 20$
- (C) $y < 2$
- (D) $y > 2$
- (E) $x = 20$

Q5. Solve the inequality $x/4 + 2 > 6$.

- (A) $x > 16$
- (B) $x < 16$
- (C) $x > 8$
- (D) $x < 8$
- (E) $x = 16$

Q6. Write the solution to $x + 1 > 9$ in interval notation.

- (A) $(8, \infty)$
- (B) $(-\infty, 8)$
- (C) $[8, \infty)$
- (D) $(-\infty, 8]$
- (E) $(-8, \infty)$

Q7. A runner runs x kilometers in a race. If she runs more than 5 kilometers in less than 30 minutes, which inequality represents this situation?

- (A) $x < 5$
- (B) $x > 5$
- (C) $x = 5$
- (D) $x \leq 5$
- (E) $x \geq 5$

Q8. Solve the inequality $3x - 2 < 14$.

- (A) $x < 16/3$
- (B) $x > 16/3$
- (C) $x < 5$
- (D) $x > 5$
- (E) $x = 16/3$

Open Ended Questions (FRQ)

Q9. A football team scores x points per game. If they score more than 10 points per game, and the score is reduced by 2 points per game due to penalties, write the inequality that represents the new score and graph the solution on a number line. Require a minimum of 3 lines to answer.

Q10. A gymnast performs a routine with x flips. If she performs more than 3 flips and each flip has a value of y points, write the inequality that represents the total score and solve for x . Assume y is a constant and x and y are positive integers. Require a minimum of 4 lines to answer.

Chef's Tips

- Read each question carefully.
- Choose the correct answer for multiple choice questions.
- Show work for free response questions.